New-16

MEMBERSHIP

Stars in catalog	44
with Gaia GSP-Phot	22 (50%)
with radial velocity	3
Stream id (sID)	46

GEOMETRY · great-circle fit

Centroid RA, Dec	188.1, -36.8 deg
Centroid l, b	298.8, 25.9 deg
Pole RA, Dec	103.6, 7.3 deg
Length (φ1, 2–98%)	22.1 deg
Width σ(φ2), robust	0.46 deg

KINEMATICS

pmRA* med ± σ	-4.16 ± 0.44 mas/yr
pmDE med ± σ	8.71 ± 1.90 mas/yr
μφ1* med ± σ	7.95 ± 1.82 mas/yr
μφ2 med ± σ	5.48 ± 0.67 mas/yr
Parallax (cat.) med	0.213 ± 0.279 mas
Parallax (wtd, ZP-corr.)	0.247 ± 0.026 mas
Distance	d = 4.1 ± 0.4 kpc
Physical length	1.57 kpc
RV median ± σ(MAD)	278.7 ± 13.7 km/s
RV gradient dV/dφ1	—

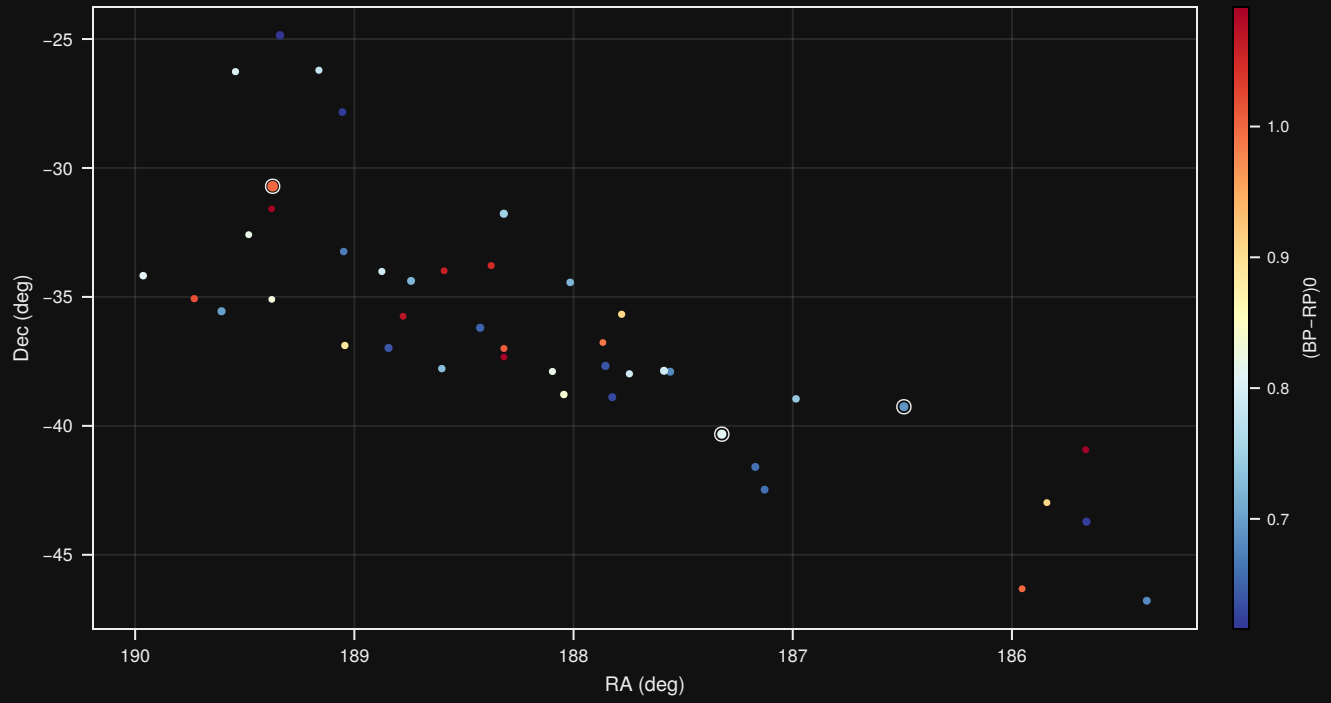
PHOTOMETRY · POPULATIONS

G0 range (median)	13.7–20.0 (18.8)
(BP−RP)0 median	0.79
GSP-Phot [M/H] (n=22)	-2.06 ± 0.86
GSP-Phot Teff median	5745 K
RUWE > 1.4	0.0 %
PM clusters (largest)	4 (10 stars)

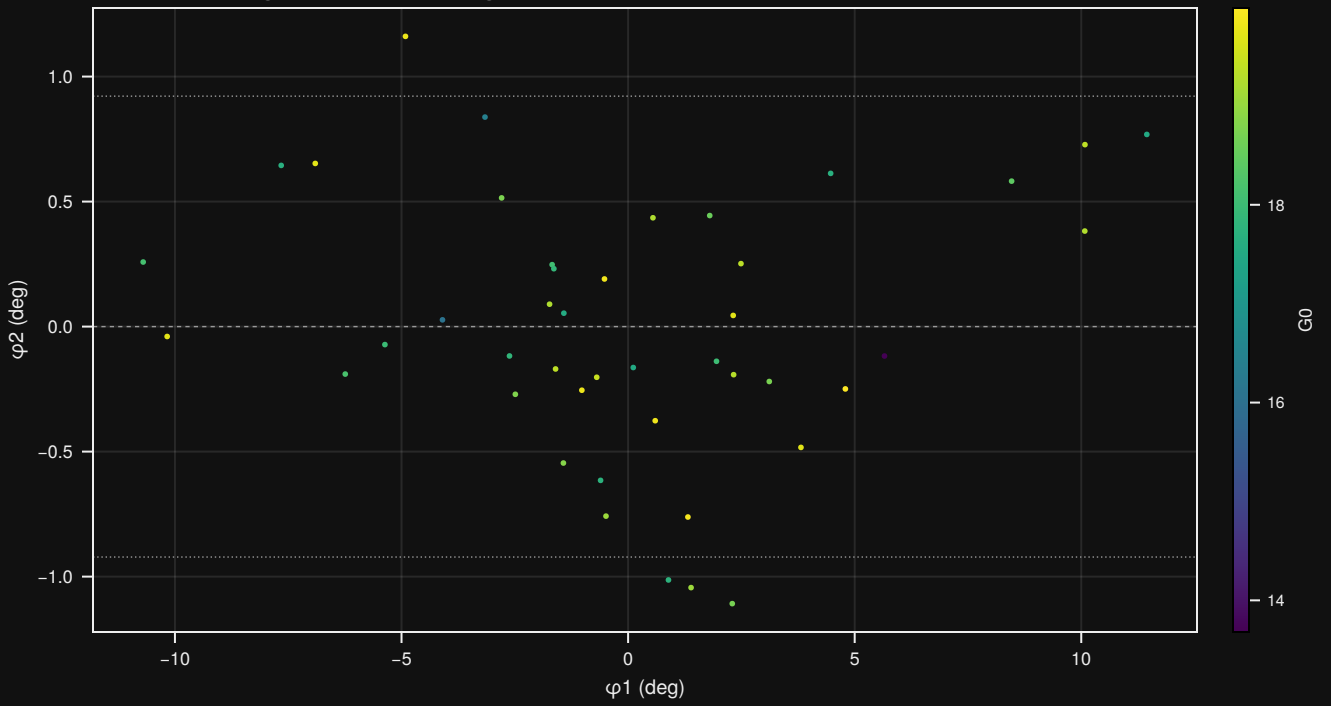
IBATA+2024 · TABLE 3

n catalog | paper	44 | 44
n_RV catalog | paper	3 | 3
<[Fe/H]> paper	-1.62
Stellar mass (paper)	170 Msun
Pole offset vs paper	1.9 deg

Sky map · catalog members (n = 44) · size ∝ brightness · ring = has RV

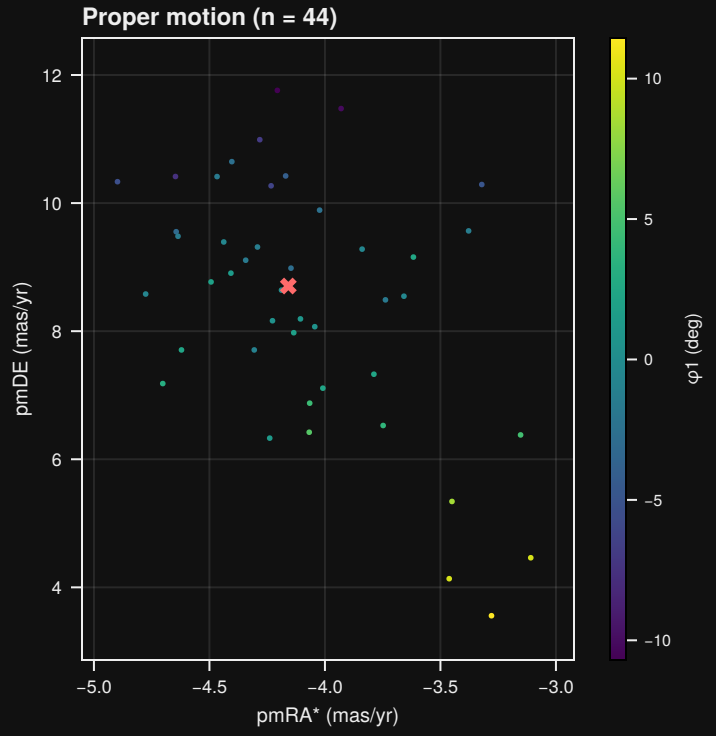


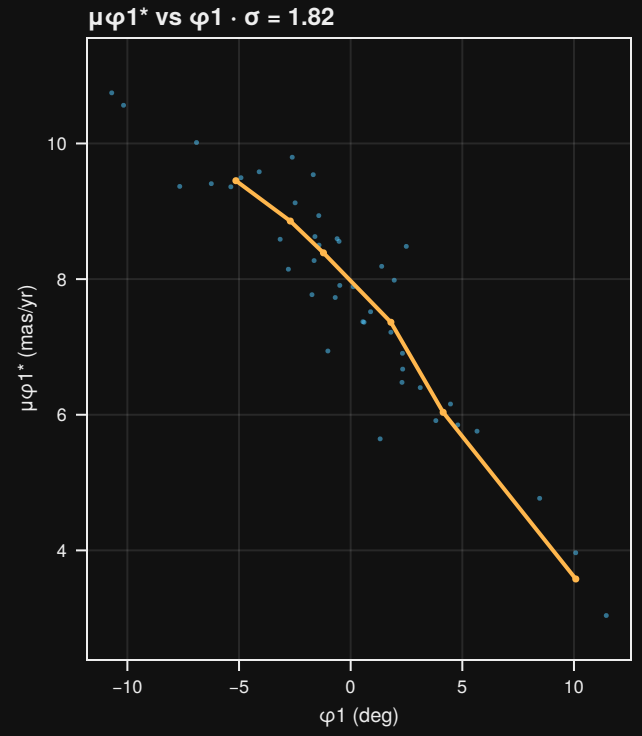
Stream frame (great-circle fit) · length 22.1° · width σ 0.46°

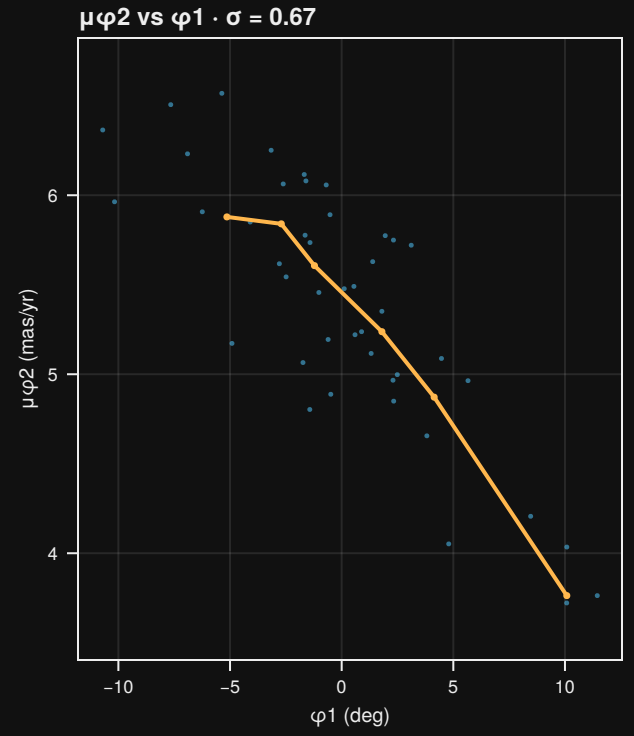


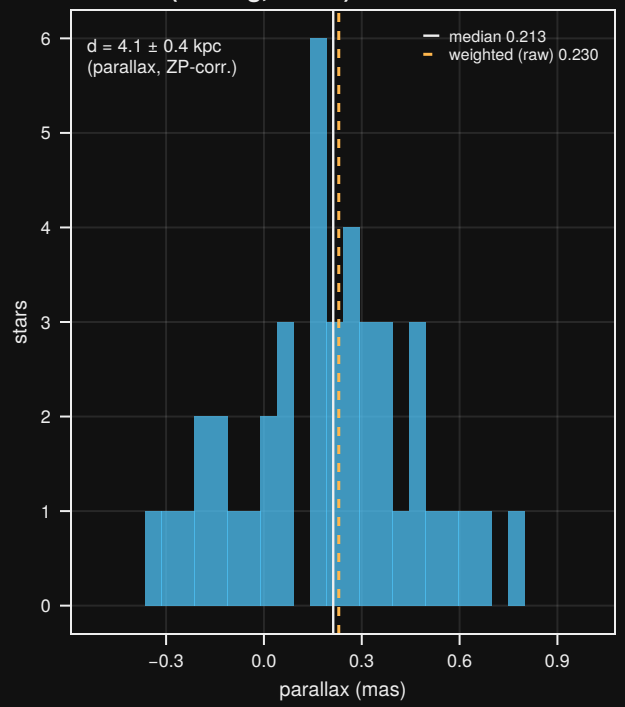
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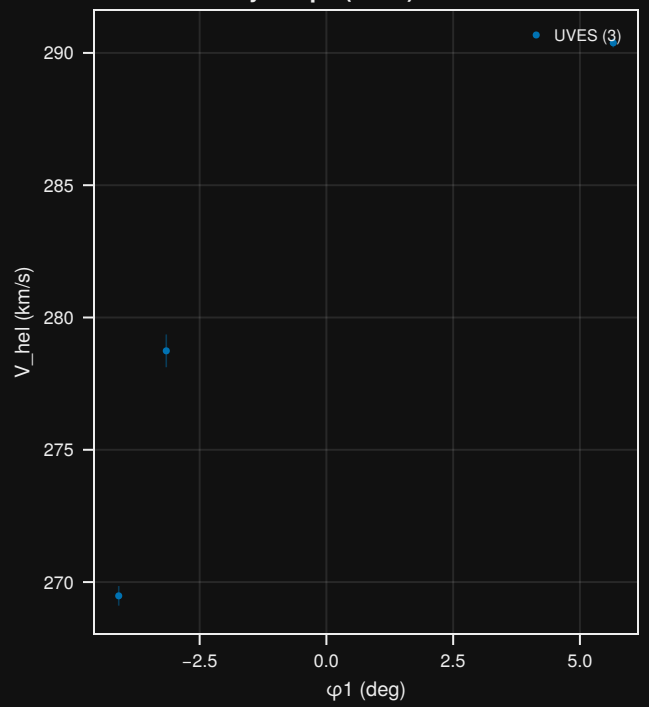
Kinematics · proper motion · parallax · radial velocity

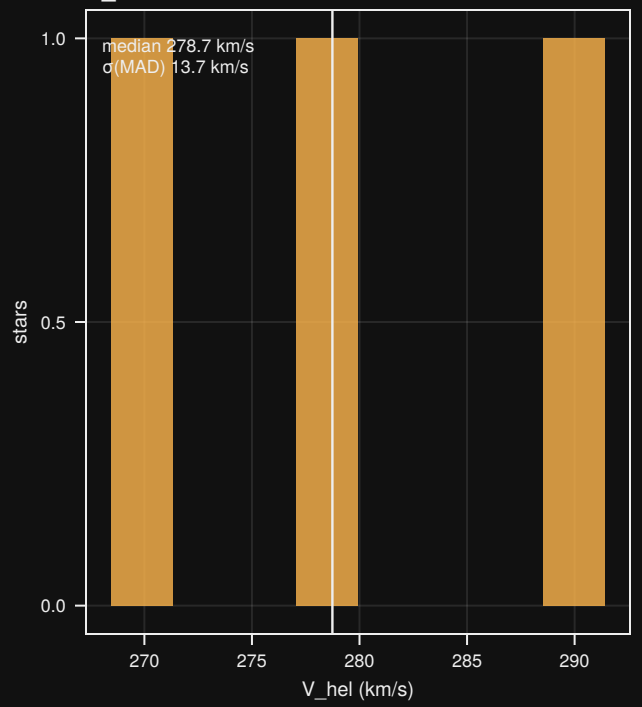






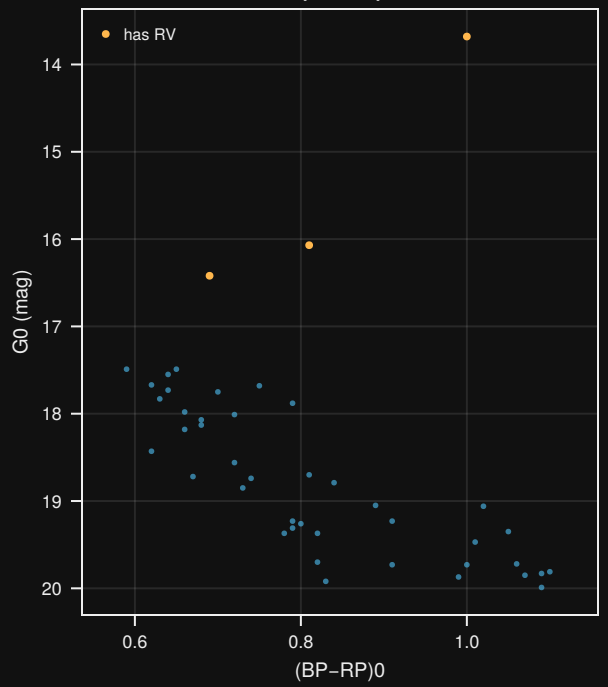




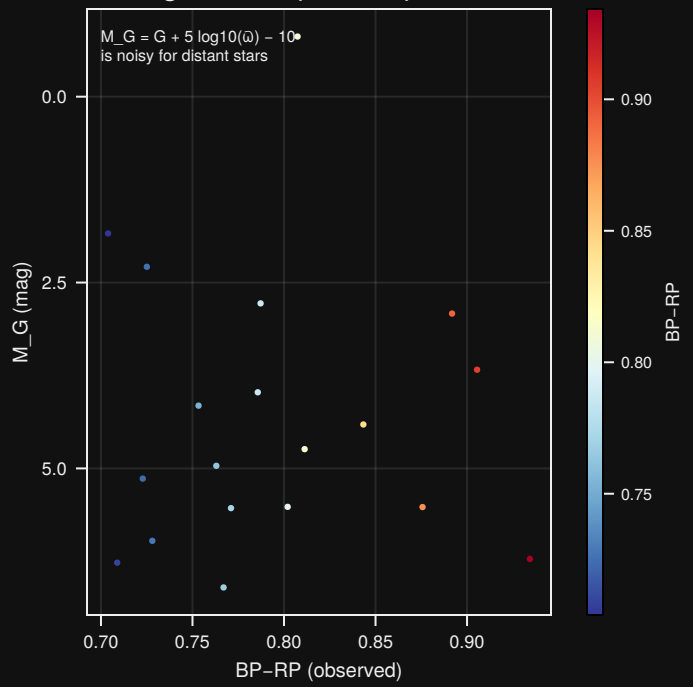


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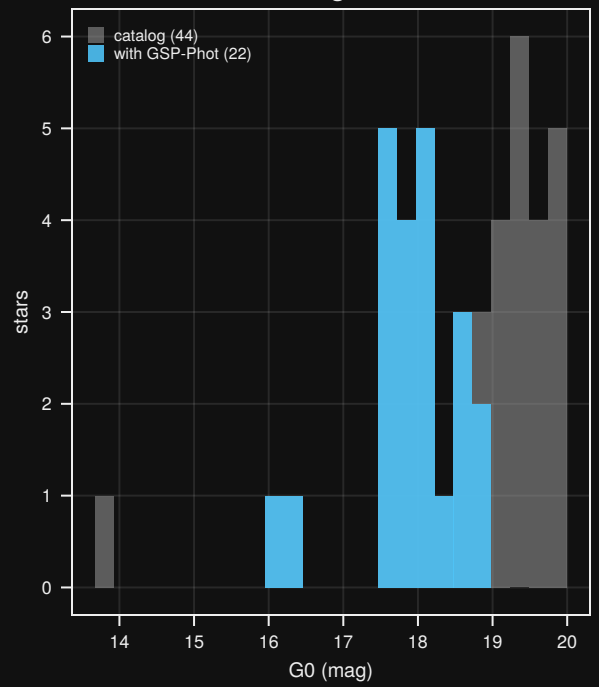
CMD · dereddened (n = 44)



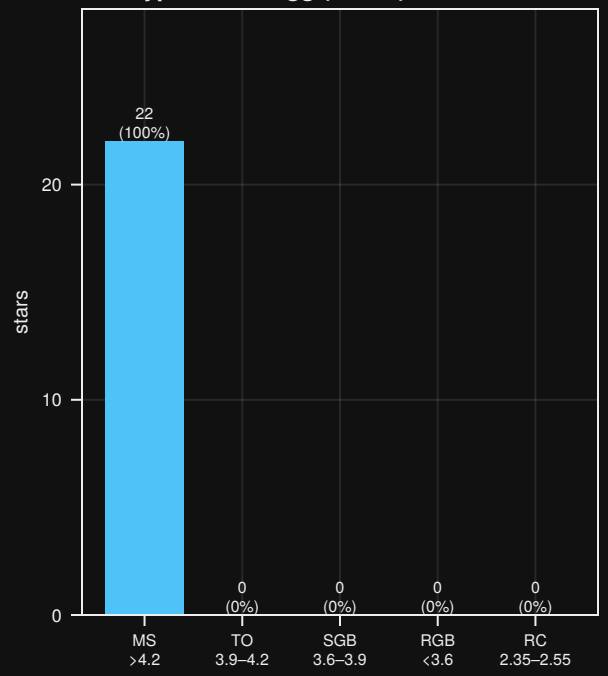
HR diagram · ϖ>0 (n = 19/22)



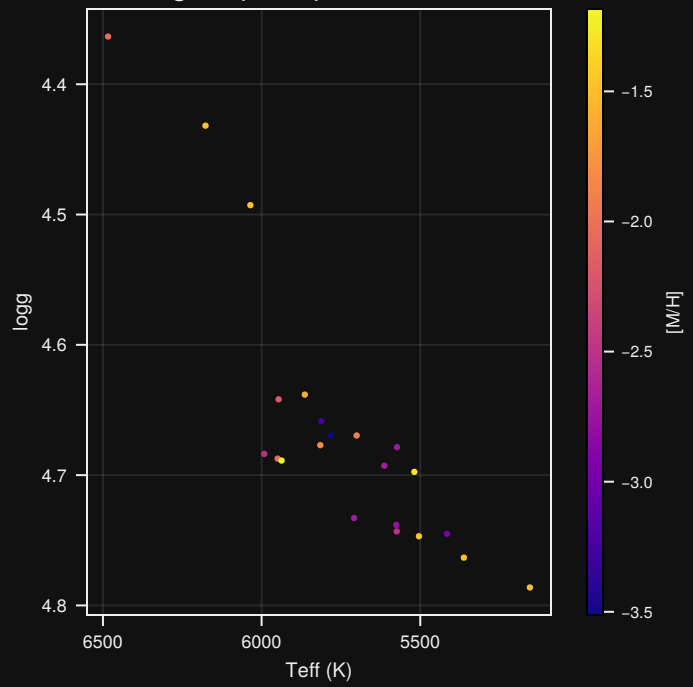
G0 & GSP-Phot coverage



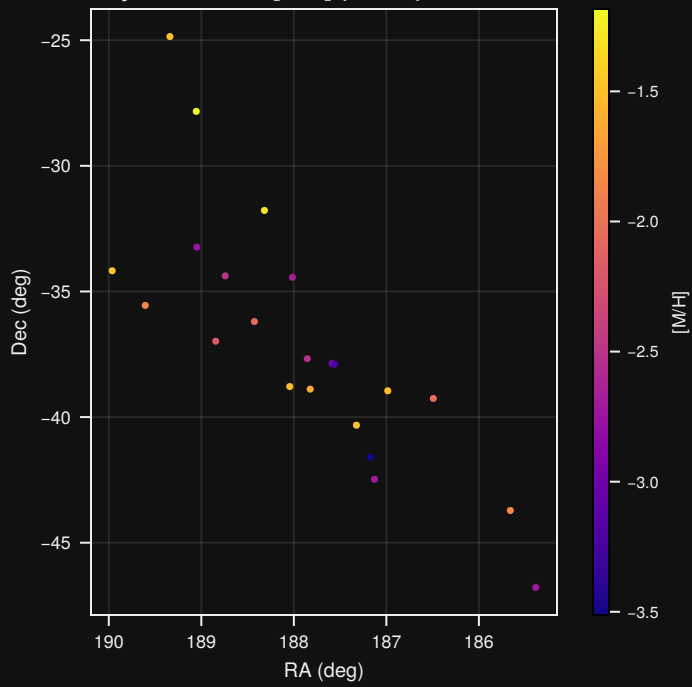
Star types from logg (n = 22)



Kiel diagram (n = 22)



Sky · GSP-Phot [M/H] (n = 22)

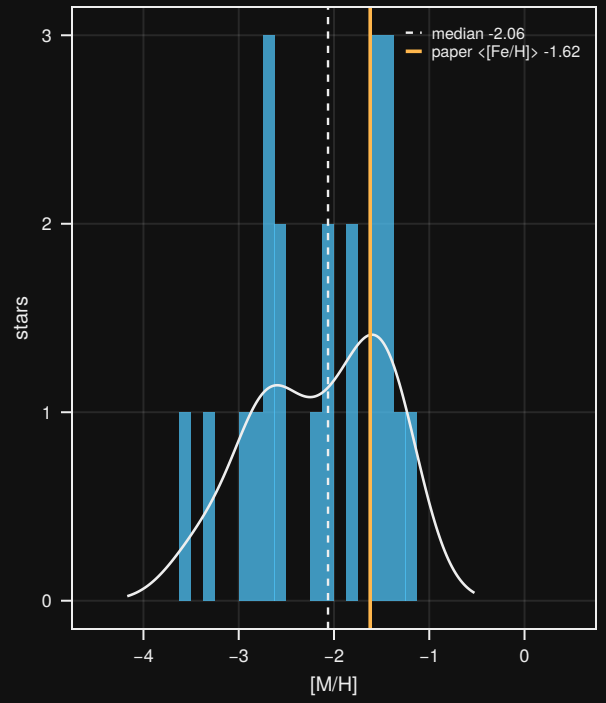


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

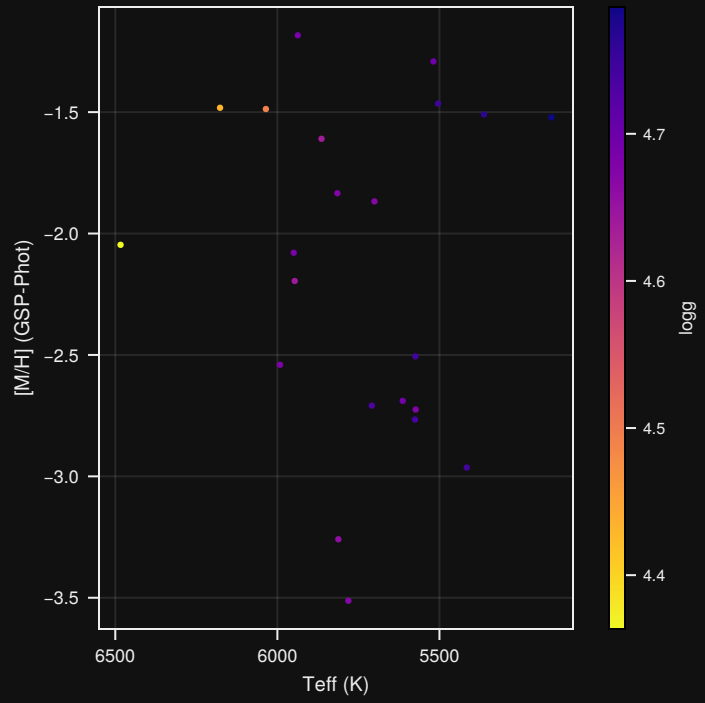
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Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 22)



Teff vs [M/H] (n = 22)



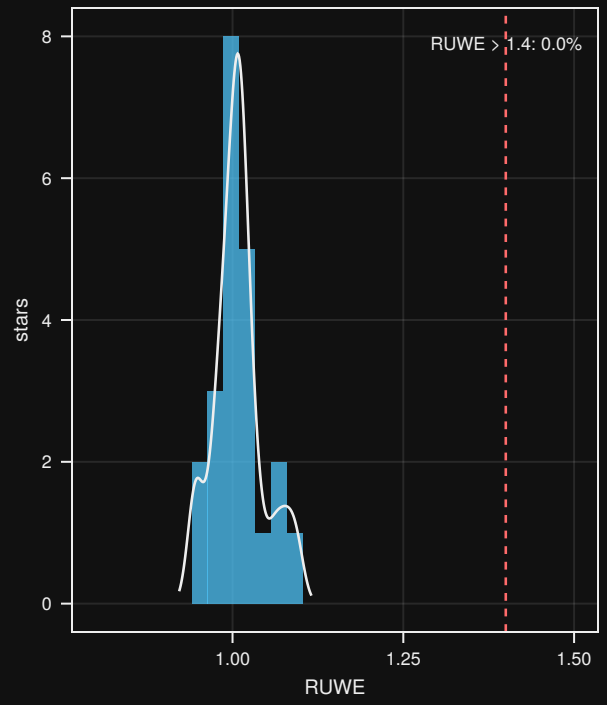
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

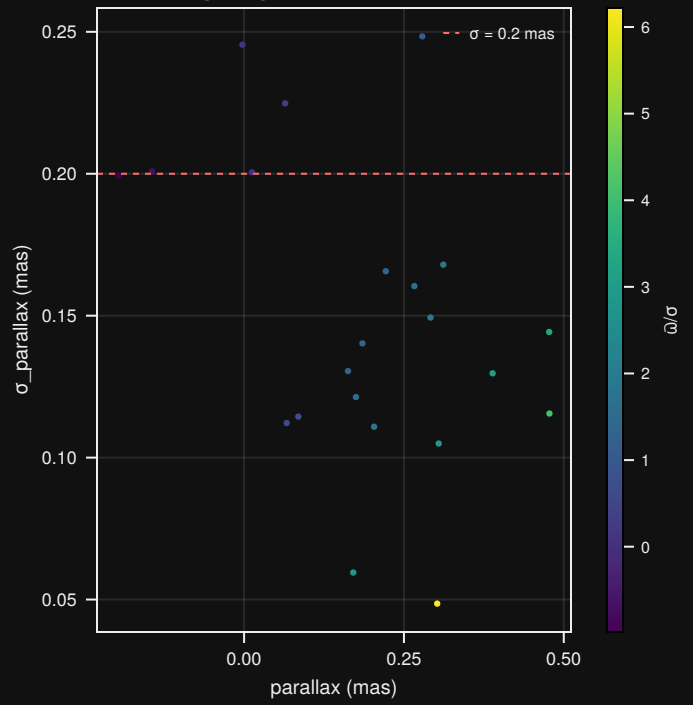
Age–metallicity (n = 0)

n < 3

RUWE (n = 22)

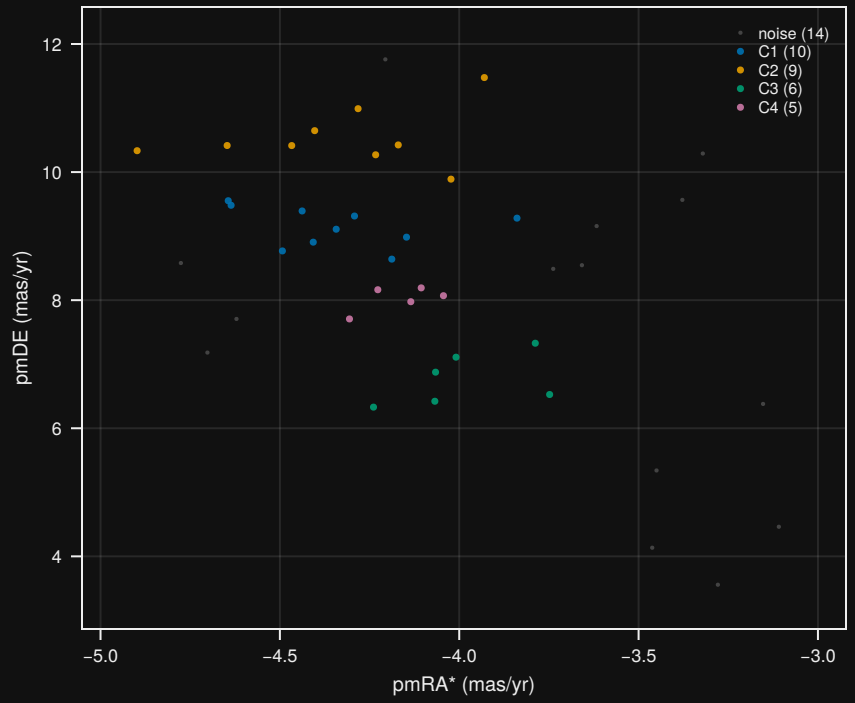


Parallax quality (n = 22)

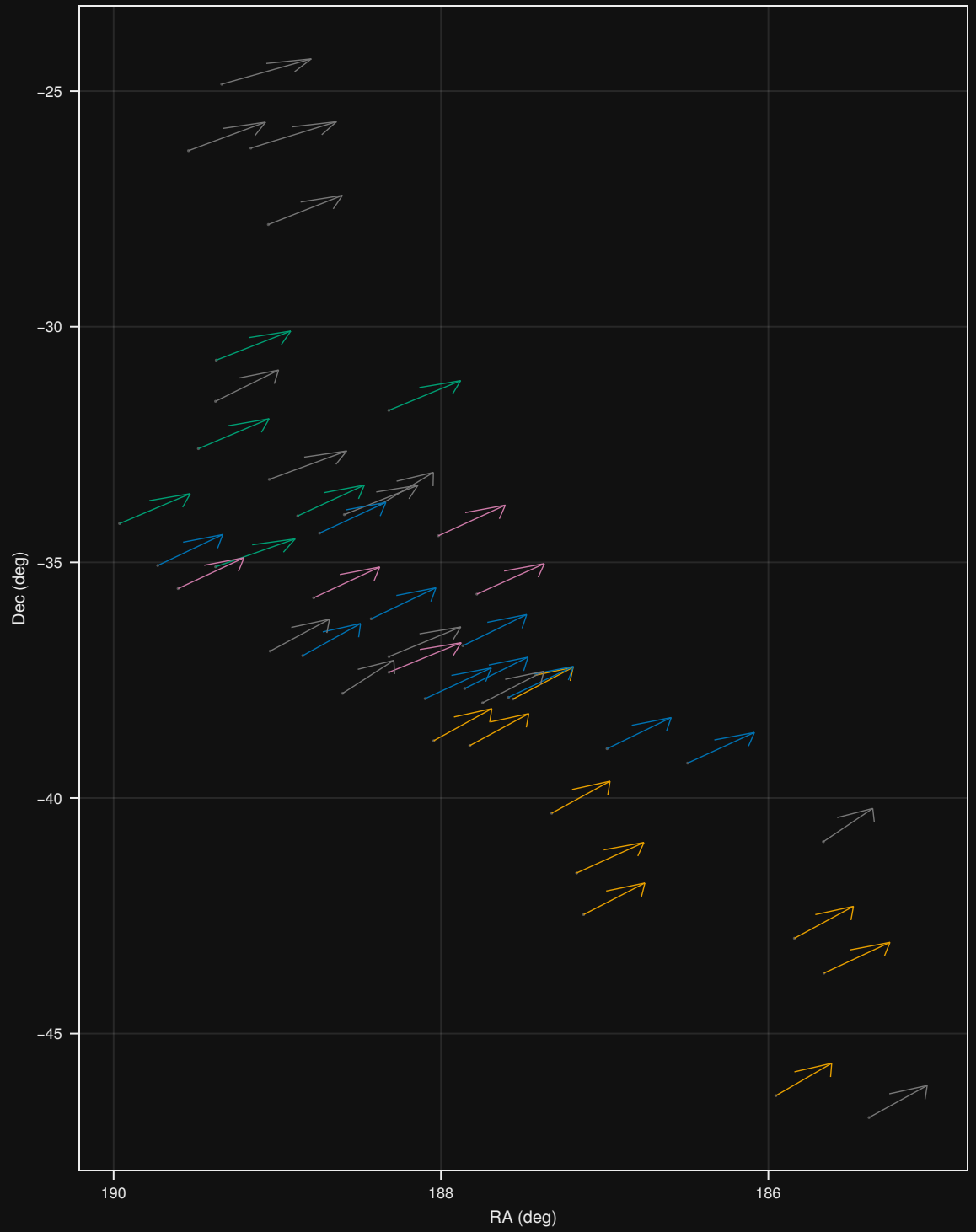


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PM clustering · HDBSCAN · 4 cluster(s) · noise 32%



Proper-motion directions (arrows) coloured by cluster



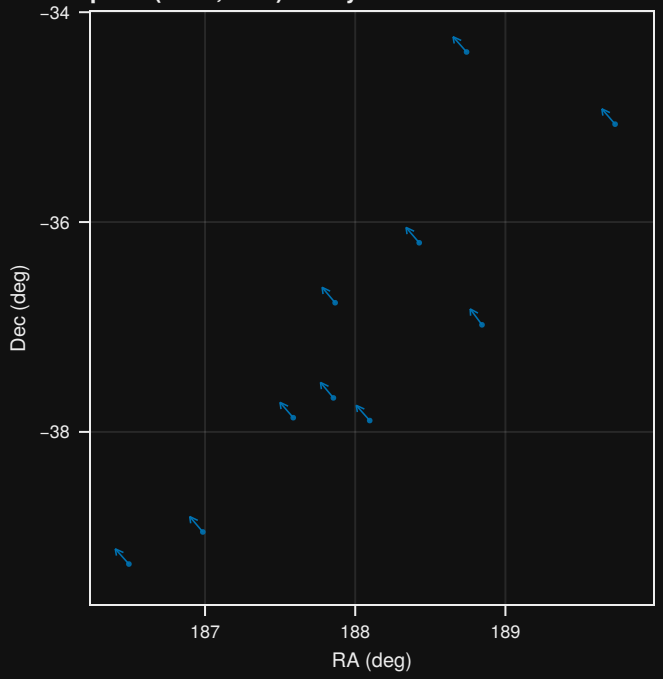
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	14	-3.54	8.10	0.36	-2.11	0
C1	10	-4.38	9.19	0.23	-2.20	1
C2	9	-4.28	10.41	0.27	-1.83	1
C3	6	-4.04	6.70	0.17	-1.38	1
C4	5	-4.13	8.07	0.13	-2.28	0

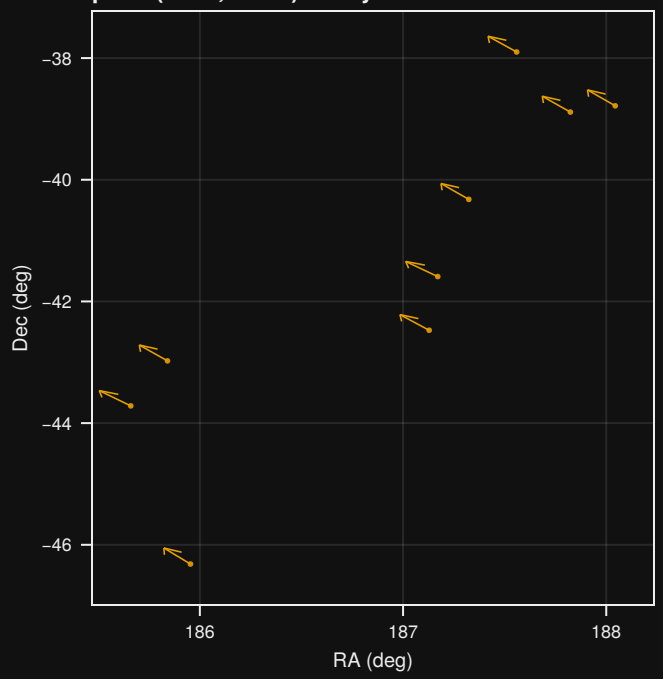
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Per-cluster proper-motion vector maps

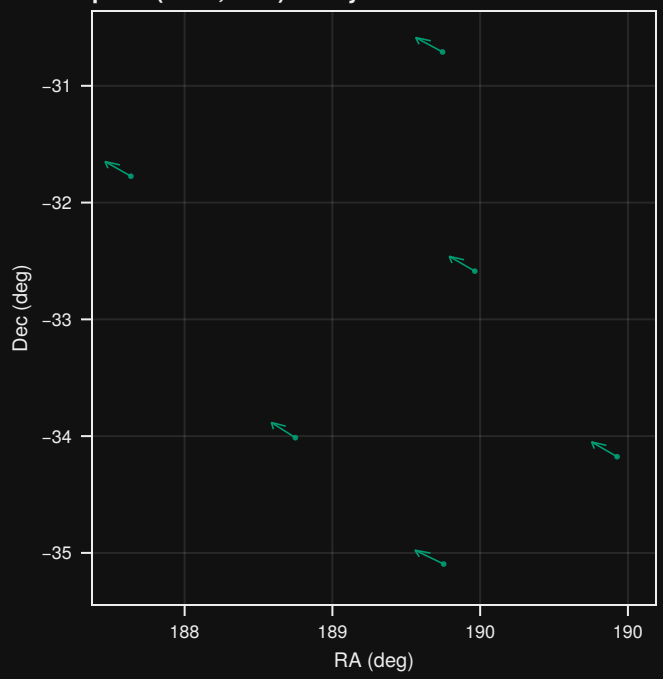
Cluster C1 · n = 10
pm = (-4.38, 9.19) mas/yr



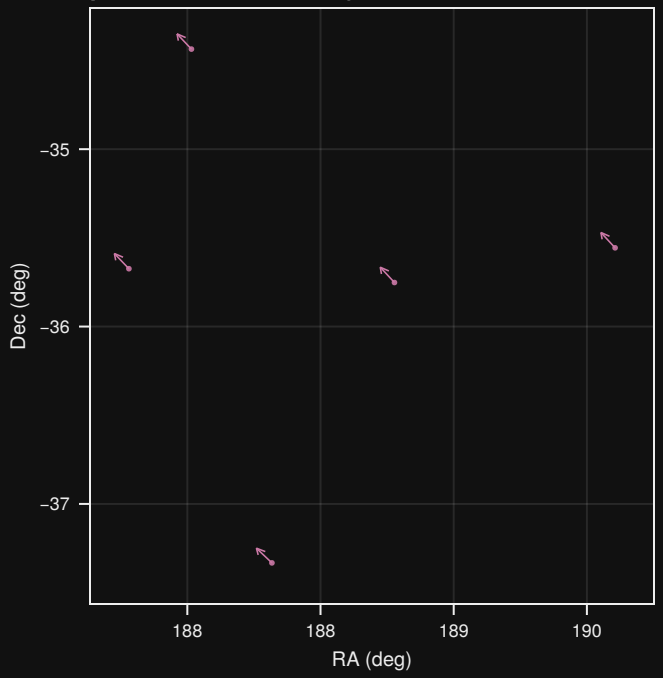
Cluster C2 · n = 9
pm = (-4.28, 10.41) mas/yr



Cluster C3 · n = 6
pm = (-4.04, 6.70) mas/yr



Cluster C4 · n = 5
pm = (-4.13, 8.07) mas/yr



Noise · n = 14

